Cell Tuition
1-1 PRIVATE TUTORS

NUMBER

Percentages and Scaling

Name _____ Date _____ No calculator Version 1

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KEY WORDS percentage *(ōrau)* · per cent *(ōrau)* · of · the whole (100%) · scale up · scale down

A Common percentages ●○○

Remember: 50% is a half, 25% is a quarter and 10% is a tenth. So 50% of 60 is $60 \div 2 = 30$ and 10% of 60 is $60 \div 10 = 6$.

1. Work out.

a) 50% of 60 = _____ **b)** 50% of 18 = _____ **c)** 10% of 70 = _____ **d)** 10% of 250 = _____

e) 25% of 80 = _____ **f)** 25% of 40 = _____ **g)** 100% of 46 = _____ **h)** 50% of 500 = _____

i) 10% of 30 = _____ **j)** 25% of 12 = _____ **k)** 50% of 7 = _____ **l)** 10% of 400 = _____

m) 25% of 200 = _____ **n)** 50% of 90 = _____ **o)** 10% of 1,000 = _____ **p)** 25% of 36 = _____

B More percentages ●●○

Remember: Use the ones you know. 20% is double 10%. 75% is three lots of 25%. 5% is half of 10%.

2. Work out.

a) 20% of 50 **b)** 75% of 40 **c)** 5% of 80 **d)** 30% of 60

e) 20% of 35 **f)** 75% of 12 **g)** 5% of 200 **h)** 70% of 30

3. Work out.

a) 50% of $30 **b)** 10% of $80 **c)** 25% of 60 kg **d)** 10% of 250 mL

C Finding the whole ●●●

Watch out: Work back to 100%. If 50% of a number is 14, the whole number is $14 \times 2 = 28$. It is not half of 14.

4. Find the number.

a) 50% of a number is 14 **b)** 10% of a number is 7 **c)** 25% of a number is 9

d) 50% of a number is 35 **e)** 10% of a number is 12 **f)** 25% of a number is 20

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D Scaling up ●●○

> **Remember:** Work out how many groups there are, then multiply. If 2 eggs make 6 muffins, then 12 eggs is 6 groups of 2 eggs, so they make $6 \times 6 = 36$ muffins.

5. Answer each question.

a) 2 eggs make 6 muffins. How many muffins can be made with 12 eggs?

b) 3 pens cost $5. How much do 12 pens cost?

c) For every 2 adults on a trip there are 10 children. There are 8 adults. How many children are there?

d) 5 tickets cost $40. How much do 15 tickets cost?

e) A tap fills 3 buckets in 4 minutes. How long does it take to fill 12 buckets?

E Scaling a recipe ●●●

> **Remember:** For twice as many people, double every amount. For half as many people, halve every amount.

6. A recipe for 4 people uses 200 g of flour, 2 eggs and 300 mL of milk.

a) How much flour is needed for 8 people?

b) How many eggs are needed for 12 people?

c) How much milk is needed for 2 people?

d) How much flour is needed for 20 people?

F Problems ●●●

7. A test has 36 questions. Kiri gets 50% of them right. How many questions does she get right?

8. Sam saves 10% of the $60 he earns. How much does he save?

9. 15 students in a class walk to school. This is 50% of the class. How many students are in the class?

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10. Find 15% of 80 by adding 10% and 5%.

11. 4 apples cost $6. How much do 10 apples cost? Find the cost of one apple first.

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1-1 PRIVATE TUTORS

NUMBER

Percentages and Scaling

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS percentage *(ōrau)* · per cent *(ōrau)* · of · the whole (100%) · scale up · scale down

A Common percentages ●○○

Remember: 50% is a half, 25% is a quarter and 10% is a tenth. So 50% of 60 is $60 \div 2 = 30$ and 10% of 60 is $60 \div 10 = 6$.

1. Work out.

a) 50% of 80 = _____ **b)** 50% of 14 = _____ **c)** 10% of 90 = _____ **d)** 10% of 350 = _____

e) 25% of 60 = _____ **f)** 25% of 20 = _____ **g)** 100% of 52 = _____ **h)** 50% of 300 = _____

i) 10% of 50 = _____ **j)** 25% of 16 = _____ **k)** 50% of 9 = _____ **l)** 10% of 600 = _____

m) 25% of 400 = _____ **n)** 50% of 70 = _____ **o)** 10% of 2,000 = _____ **p)** 25% of 28 = _____

B More percentages ●●○

Remember: Use the ones you know. 20% is double 10%. 75% is three lots of 25%. 5% is half of 10%.

2. Work out.

a) 20% of 40 **b)** 75% of 80 **c)** 5% of 60 **d)** 30% of 70

e) 20% of 45 **f)** 75% of 20 **g)** 5% of 400 **h)** 70% of 20

3. Work out.

a) 50% of $50 **b)** 10% of $60 **c)** 25% of 40 kg **d)** 10% of 750 mL

C Finding the whole ●●●

Watch out: Work back to 100%. If 50% of a number is 14, the whole number is $14 \times 2 = 28$. It is not half of 14.

4. Find the number.

a) 50% of a number is 17 **b)** 10% of a number is 9 **c)** 25% of a number is 7

d) 50% of a number is 45 **e)** 10% of a number is 15 **f)** 25% of a number is 30

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D Scaling up ●●○

> **Remember:** Work out how many groups there are, then multiply. If 2 eggs make 6 muffins, then 12 eggs is 6 groups of 2 eggs, so they make $6 \times 6 = 36$ muffins.

5. Answer each question.

a) 2 eggs make 8 muffins. How many muffins can be made with 10 eggs?

b) 4 pens cost $6. How much do 12 pens cost?

c) For every 2 adults on a trip there are 12 children. There are 6 adults. How many children are there?

d) 5 tickets cost $30. How much do 20 tickets cost?

e) A tap fills 2 buckets in 3 minutes. How long does it take to fill 12 buckets?

E Scaling a recipe ●●●

> **Remember:** For twice as many people, double every amount. For half as many people, halve every amount.

6. A recipe for 6 people uses 300 g of flour, 3 eggs and 600 mL of milk.

a) How much flour is needed for 12 people?

b) How many eggs are needed for 18 people?

c) How much milk is needed for 3 people?

d) How much flour is needed for 2 people?

F Problems ●●●

7. A test has 36 questions. Kiri gets 25% of them right. How many questions does she get right?

8. Sam saves 10% of the $80 he earns. How much does he save?

9. 12 students in a class walk to school. This is 50% of the class. How many students are in the class?

★ Extension

10. Find 15% of 60 by adding 10% and 5%.

11. 6 apples cost $9. How much do 10 apples cost? Find the cost of one apple first.

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